

4(a)	any five from: 1 gene / DNA, from <i>Bacillus thuringiensis</i> ; 2 ref. to <u>restriction</u>, enzyme / endonuclease ; 3 clone gene / amplify gene, using PCR ; 4 insert, gene / DNA, into plasmid ; 5 add promoter ; 6 seal (plasmid) using DNA <u>ligase</u> ; 7 (maize) tissue / cell / callus / embryo, takes up plasmid / recombinant DNA ; 8 (maize) expresses new gene / produces Bt toxin ; 9 AVP ; e.g. Ti (plasmid) / <i>Agrobacterium tumefaciens</i>	5
4(b)(i)	any two from: 1 pollen from Bt maize, kills / is toxic to / harms, caterpillars ; 2 pollen from non-Bt maize, does not kill / is not toxic to / does not harm, caterpillars ; 3 (eating) leaves with no pollen does not kill / is not toxic to / does not harm, caterpillars ; 4 more days of eating pollen from Bt maize gives more caterpillar deaths / AW ;	2

Question	Answer	Marks
4(b)(ii)	<i>any four from:</i> 1 Bt pollen did not get deposited (onto milkweed plants) / AW ; 2 caterpillars and Bt maize located in different areas ; 3 pollen released at a different time to when caterpillars feed ; 4 <i>ref. to</i> caterpillars develop resistance (to Bt toxin) ; 5 <i>ref. to</i> lab and, quantity / toxicity, of Bt pollen ; 6 AVP ; e.g. in 2019 breeding conditions were better than in 1999 e.g. herbicides may reduce milkweed availability in maize fields	4